



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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COMMON QUARTERLY EXAMINATION - 2024 - 25

Time Allowed : 3.00 Hours]

BIOLOGY

[Max. Marks : 70

Part - I (Bio-Botany) (Marks : 35)

Section - I

Note. (i) Answer All the questions.

8×1=8

(ii) Choose the most appropriate answer from the given four alternatives and write the option code and corresponding answer.

- The correct Statement regarding Blue green Algae is
 - lack of motile structures
 - Presence of cellulose in cell wall
 - absence of mucilage around the thallus
 - presence of floridean starch
- Which of the following represents gametophytic generation in pteridophytes?
 - Prothallus
 - Thallus
 - Cone
 - Rhizophore
- Select the Mismatch pair
 - Musa - Unicostate
 - Lalab - Trifoliate
 - Acalypha - Leaf mosaic
 - Allamanda - Ternate Phyllotaxy
- Gynoecium with united carpels is termed as
 - Apocarpous
 - Multicarpellary
 - Syncarpous
 - None of the above
- In spase of the cell cycle
 - Amount of DNA doubles in each cell
 - Amount of DNA remains same in each cell
 - Chromosome number is increased
 - Amount of DNA is reduced to half in each cell
- The β -D Glucose units in cellulose are linked together by
 - N - acetyl side chains
 - N - acetyl D - glucosamine
 - 1→3 Linkage
 - β - (1,4) glycosidic linkage
- Match the Column and identify the correct option.

Column - I	Column - II	A	B	C	D
A Carolus Linnaeus	(i) Papilionaceous corolla	a) 1	4	3	2
B Fabaceae	(ii) Introduced Binomial	b) 3	1	2	4
C Cladogram	(iii) Artificial system of classification	c) 3	2	4	1
D Gaspard Bauhin	(iv) Tree - shaped diagram	d) 3	1	4	2
- Assertion (A) : Mitochondria is considered as semi-autonomous organelle as it contains DNA
Reason (R) : Because they are inherited from male parent only
 - Both Assertion and Reason are true
 - Both Assertion and Reason are false
 - Assertion is true Reason is False
 - Assertion is false but Reason is true

SECTION - II

Answer any four of the following.

4×2=8

- Why Viruses are called 'Biologist Puzzle'?
- Mention two Characters shared by Gymnosperms & Angiosperms.
- What are Multiple fruit? Give an example.
- Write the Significance of DNA bar coding.
- "Meiosis is otherwise called & educational division" - Justify your answer.
- Define Enzymes.

SECTION - III

Answer any three questions. Question No. 19 is compulsory.

3×3=9

- Write the distinguishing features of Monera
- What are mixed inflorescence? Explain its types.
- Write the Properties of water.
- Draw a neat labelled diagram of chloroplast structure.
- Differentiate between Taxonomy & Systematic.

SECTION - IV

Answer all the questions.

2×5=10

- (a) Explain the structure of DNA
(OR)
(b) Briefly describe the various stages of mitosis and a division
- (a) Write the Botanical description of Datura Metel with an floral diagram & formula
(OR)
(b) Explain any 5 types of stele with an diagram.



PART - II (BIO-ZOOLOGY) (Max. Marks: 35)

SECTION - I

I. Choose the correct answer.

8x1=8

1. The book "Historia Generalis Plantarum" was written by
a) Aristotle b) De Candolle c) John Ray d) Walter Rosen
2. Excretion is carried out by ----- in Cockroach.
a) Flame cells b) Kidney c) Nephredia d) Malpighian Tubules
3. ----- Produces heat by non-shivering thermogenesis in neonates.
a) Brown fat b) Collagen c) Reticular fibres d) White fat
4. **Assertion [A]** : Only 10% of Oxygen is transported in dissolved form by blood.
Reason [R] : Because, Oxygen is poorly soluble in water.
a) A and R are correct. and R is the correct explanation for A.
b) A and R are correct. But R is not the correct explanation for A.
c) A is correct, but R is wrong d) A is wrong, but R is correct
5. Blood group is due to the presence or absence of surface -----
a) Antigens on the surface of WBC b) Antibodies on the surface of RBC
c) Antigens on the surface of RBC d) Antibodies on the surface of WBC
6. Hormone which helps in the reabsorption of water in kidney tubules is
a) cholecystokinin b) angiotensin II c) antidiuretic hormone d) pancreozymin
7. Match Column - I with Column - II and choose the correct options.

Column - I	Column - II	
(P) Small intestine -	(i) Largest factory	a) (P-iv); (Q-iii), (R-i); (S-ii)
(Q) Pancreas -	(ii) Absorption of Water	b) (P-iii); (Q-ii), (R-i); (S-iv)
(R) Liver -	(iii) Carrying electrolytic solution	c) (P-iv); (Q-iii), (R-ii); (S-i)
(S) Colon -	(iv) Digestion and absorption	d) (P-ii); (Q-iv), (R-iii); (S-i)
8. Which of the following is a crustacean?
a) Prawn b) Snail c) Sea anemone d) Hydra

SECTION - II

II. Answer any Four of the following in one or two sentences.

4x2=8

9. Mention any 2 Molecular tools of classification.
10. Define Residual volumes.
11. What is Atherosclerosis?
12. Write down the systematic position of Cockroach.
13. What is the role of HCl secreted in the stomach in digestion?
14. Give two differences between Chondrichthyes and Osteichthyes.

SECTION - III

III. Answer any Three of the following. Q.No.19 is Compulsory.

3x3=9

15. What are Ammonotaxa? Give example.
16. What are called living fabrics? Why?
17. Differentiate between Polyp and Medusa.
18. List down the causes of Peptic ulcer.
19. What are heart sounds? When and how are these sounds produced?

SECTION - IV

IV. Answer in Detail, Draw diagrams where ever necessary.

2x5=10

20. a) Members of which class possess pneumatic bones? Write down the distinguishing features of the class.

(OR)

- b) Explain the Nervous system of Frog with an appropriate diagram.
21. a) What are the events taking place in Inspiration and Expiration?

(OR)

- b) Describe the process of Glomerular Filtration in the formation of Urine.